

MQTT – Current Sensor – Module

Summary:

Module for the ESP32 S3 based modular controller board.
Module is designed to run 3 x 4–20mA Sensors, and read them
at a resolution of 15 bit.

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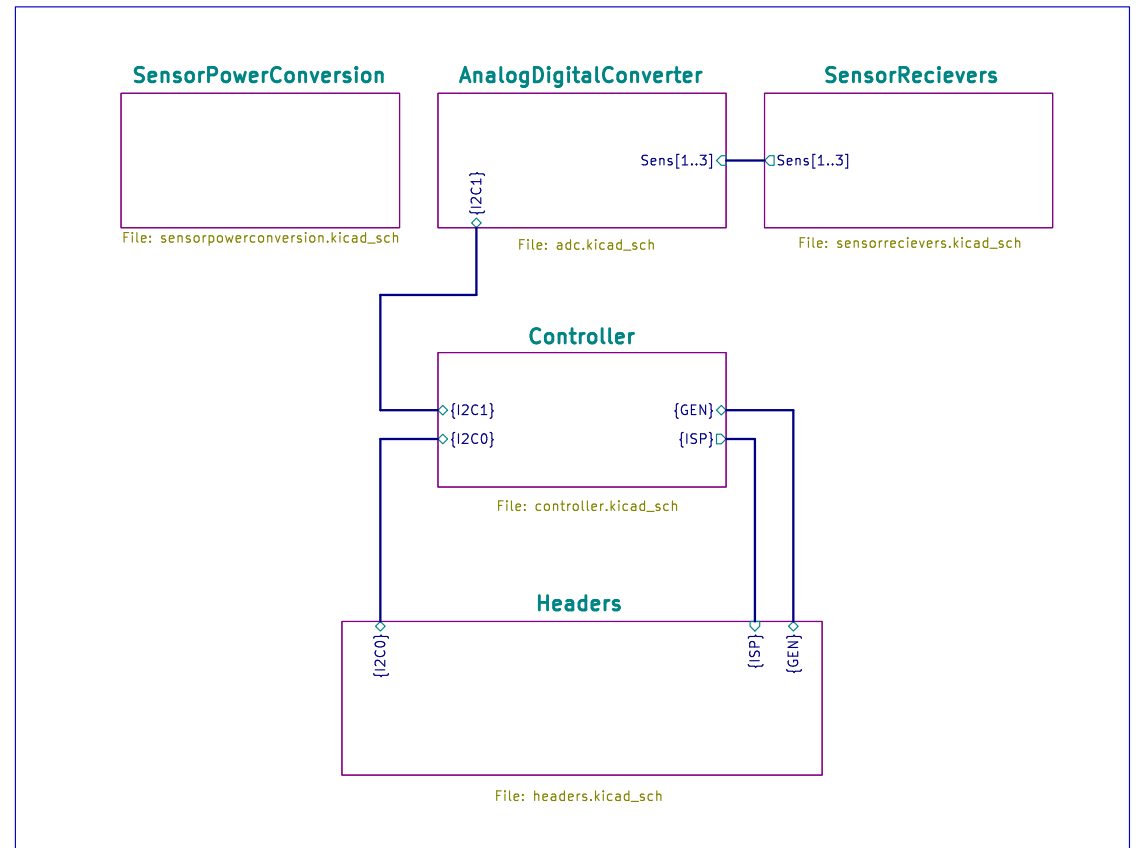
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Non BOM Items

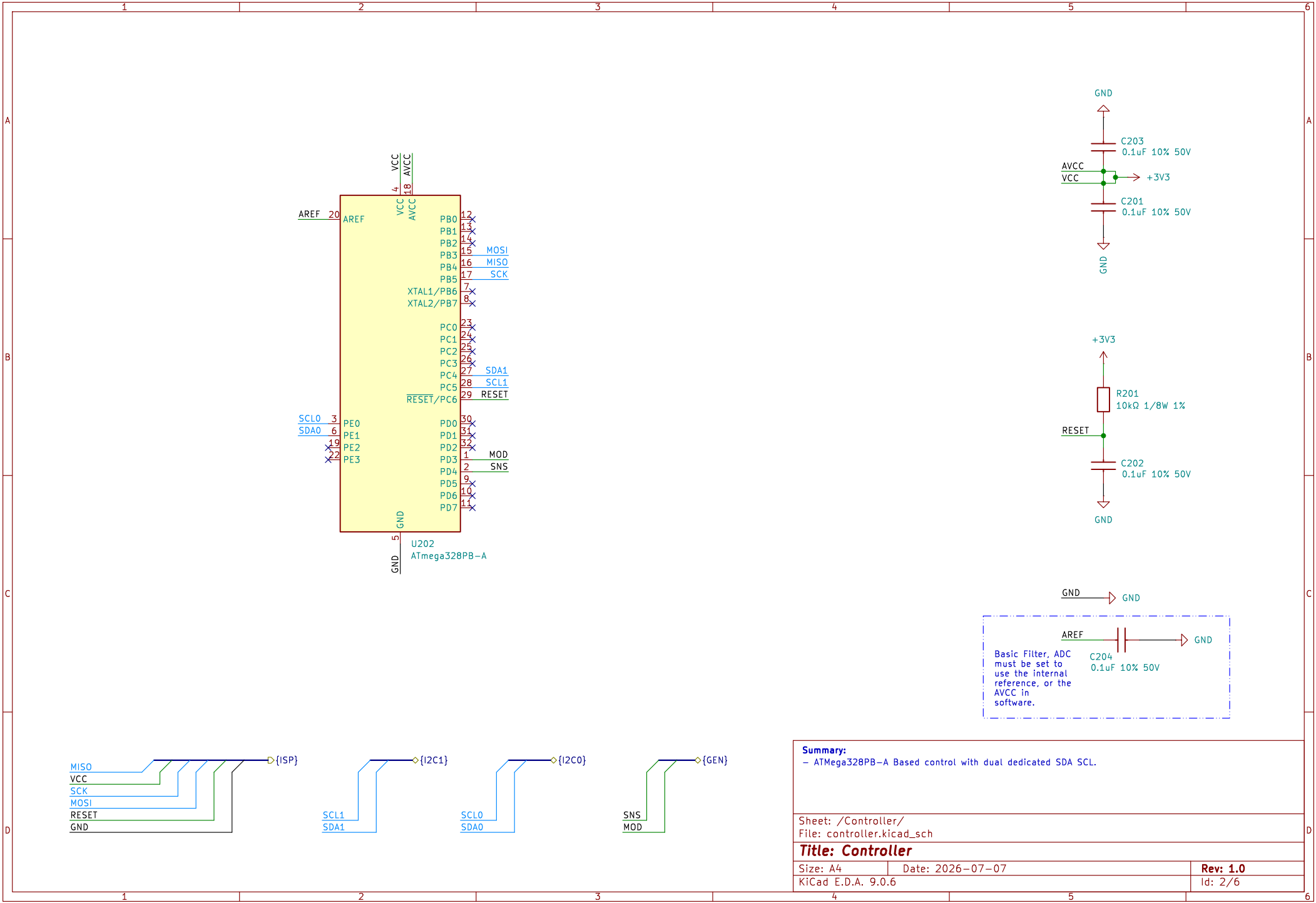
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Additional BOM Items

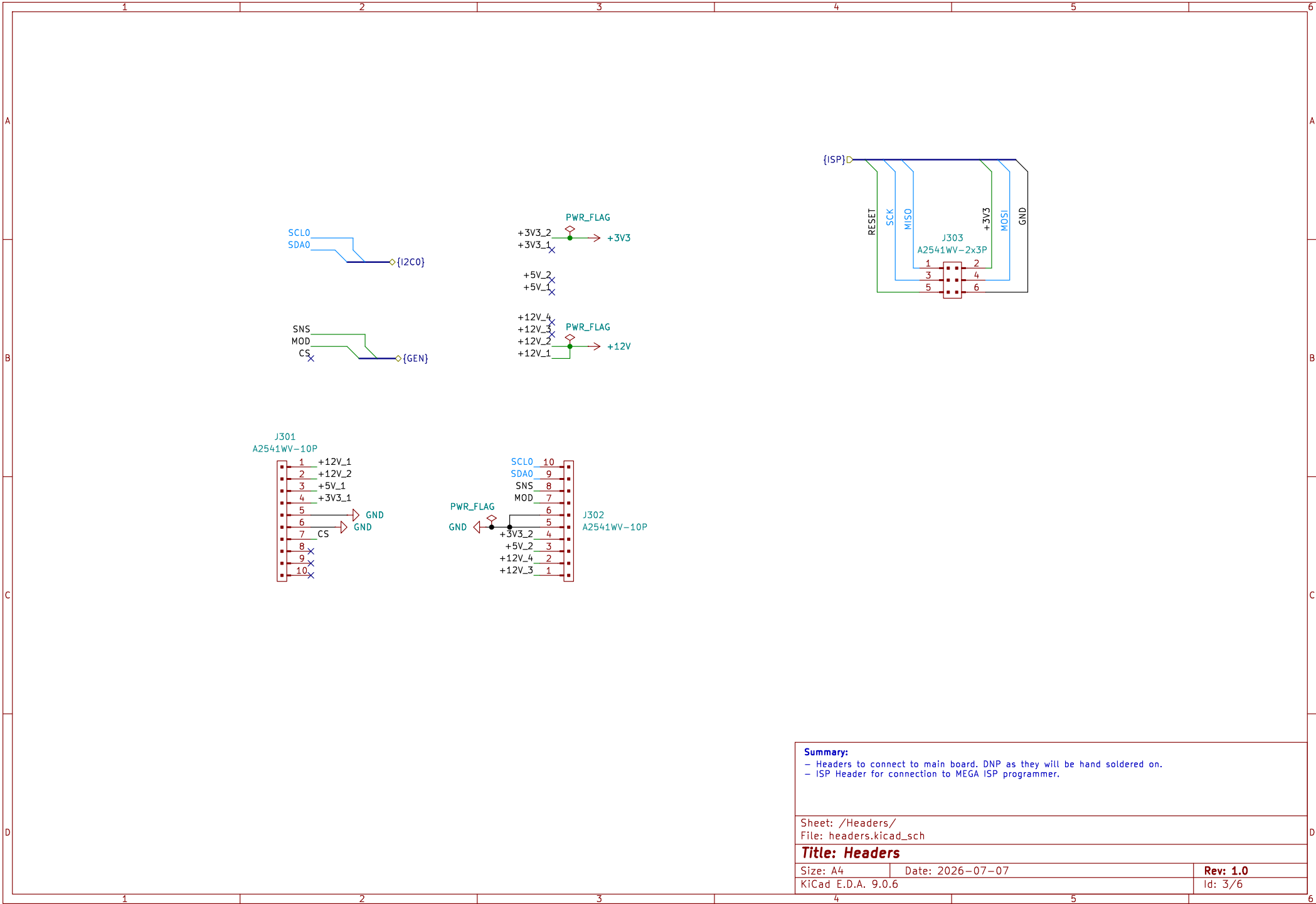
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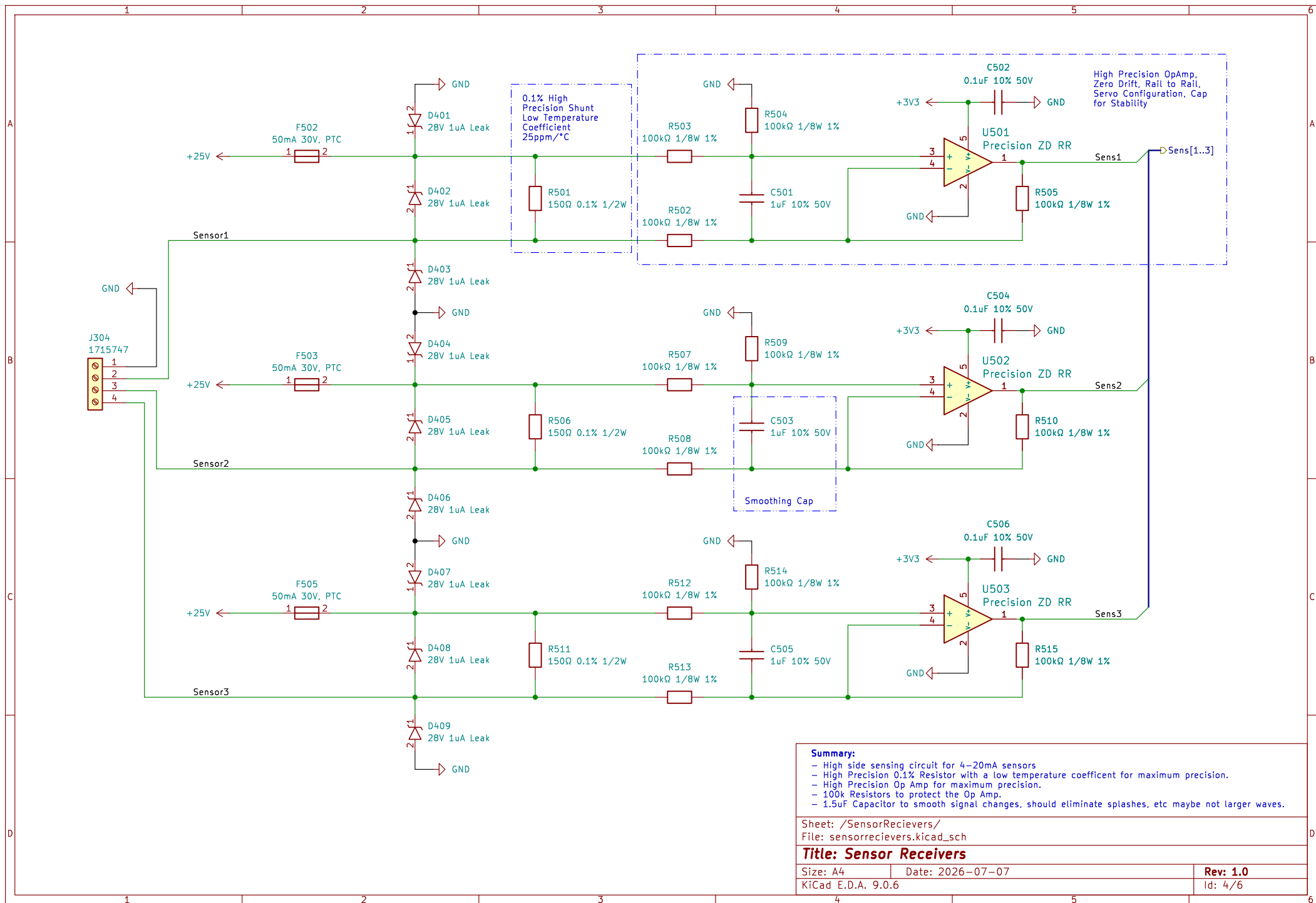


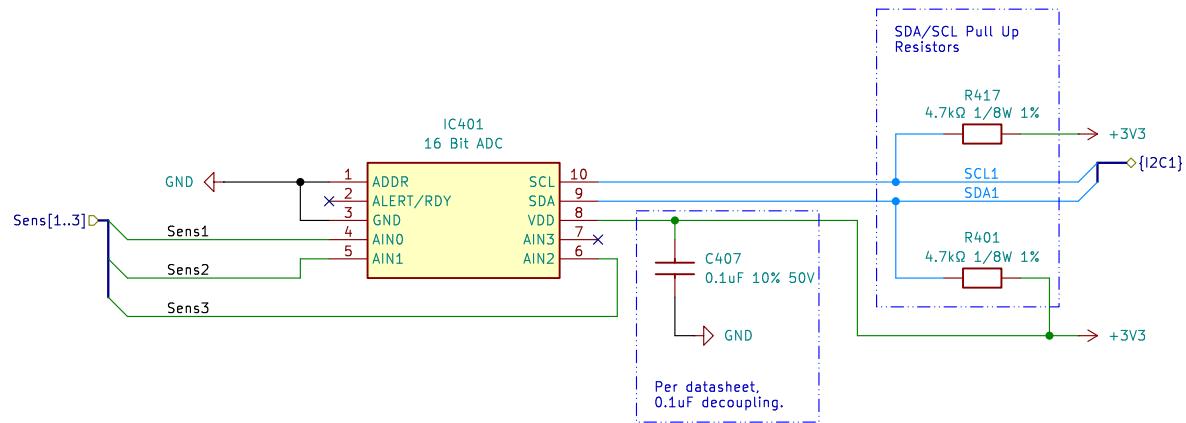
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File: MQTT–Module–CurrentSensors.kicad_sch		
Title: MQTT – Current Sensor – Module		
Size: A4	Date: 2026–07–07	Rev: 1.0
KiCad E.D.A. 9.0.6	Id: 1/6	



Summary: – ATmega328PB–A Based control with dual dedicated SDA SCL.		
Sheet: /Controller/ File: controller.kicad_sch		
Title: Controller		
Size: A4	Date: 2026–07–07	Rev: 1.0
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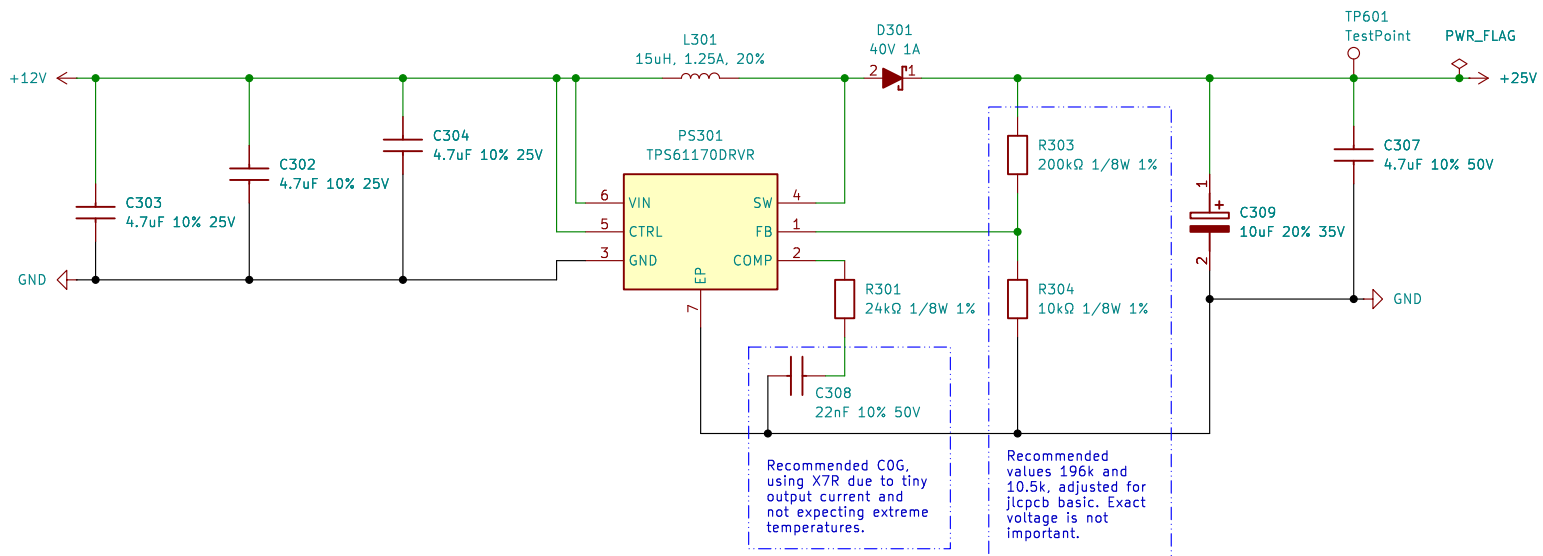


Summary:
 - ADS1115 configured to use base address (0x48).

Sheet: /AnalogDigitalConverter/
 File: adc.kicad_sch

Title: Analog Digital Converter

Size: A4	Date: 2026-07-07	Rev: 1.0
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Summary:

- Texas Instruments generated Boost Converter, 12V to -25V
- Used a shielded inductor for minimal interference.

Sheet: /SensorPowerConversion/

File: sensorpowerconversion.kicad_sch

Title: Sensor Power Conversion

Size: A4

Date: 2026-07-07

Rev: 1.0

KiCad E.D.A. 9.0.6

Id: 6/6